Blow-down to physical variables: the folded-cycle escape theorem

assembling the chart results into the original $(\varepsilon_1, \varepsilon_2)$ system

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Abstract

We carry the chart results back through the geometric blow-up to the physical variables, completing Path A's leading-order-with-prefactor analysis. The fold-weighted blow-up $r = \varepsilon_2^{1/3} R$, $y = \varepsilon_2^{2/3} Y$, $t = \varepsilon_2^{-1/3} T$ maps the physical SDE *exactly* onto the inner Riccati with effective noise $\eta = \sigma/\sqrt{\varepsilon_2}$ (an Itô change of variables, no anomaly). Blowing down the chart conclusions then gives, in the original variables: the critical-noise law $\sigma_*^A = 2\sqrt{\pi}\,\sqrt{\varepsilon_2}\,G$, the inner exit (peel-off) measure TW_β with $\beta = 4\varepsilon_2/\sigma^2$ at physical scales $(r, y) \sim (\varepsilon_2^{1/3}, \varepsilon_2^{2/3})$, the early-escape probability $1 - F_\beta(0)$, and the early-escape rate $e^{-c\,\varepsilon_2/\sigma^2}$ with $c = 5.4439\ldots$ The geometry factor G is $\sqrt{ac/b}$ frozen ($\alpha = 2$) or $\sqrt{\langle a\rangle\langle c\rangle/\langle b\rangle}$ averaged ($\alpha = 1$). The noise collapse in $\sigma/\sqrt{\varepsilon_2}$ is verified in the physical variables. This assembles the six companion notes into one statement.

1 The blow-up and the noise covariance

The JKK normal form near the folded-cycle fold is, in physical variables,

$$\mathrm{d}r = \big(b(\theta)r^2 - a(\theta)y\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \mathrm{d}y = -\varepsilon_2\,c(\theta)\,\mathrm{d}t, \qquad \dot\theta = \varepsilon_1, \tag{1}$$

with $a, b, c > 0$ smooth 1-periodic and $0 < \varepsilon_2 \ll \varepsilon_1 \ll 1$, $\varepsilon_1 = \varepsilon^\alpha$, $\varepsilon_2 = \varepsilon^3$. The Krupa–Szmolyan fold blow-up is the quasi-homogeneous (weights $(1, 2, 3)$ on (r, y, ε_2)) map covered by charts K_1 (entry), K_2 (rescaling), K_3 (exit). The central chart K_2 is $\rho = \varepsilon_2^{1/3}$:

$$r = \varepsilon_2^{1/3} R, \qquad y = \varepsilon_2^{2/3} Y, \qquad T = \varepsilon_2^{1/3} t. \tag{2}$$

Proposition 1 (exact noise covariance)**.** *Under (2), (1) becomes the inner Riccati*

$$\mathrm{d}R = \big(bR^2 - aY\big)\,\mathrm{d}T + \eta\,\mathrm{d}B_T, \qquad \mathrm{d}Y = -c\,\mathrm{d}T, \qquad \boxed{\eta = \sigma/\sqrt{\varepsilon_2}},$$

exactly (an Itô change of variables on the diffeomorphism (2); the blow-up introduces no Itô correction because it is a smooth coordinate change). After the linear rescaling absorbing a, b, c this is the canonical $\mathrm{d}R = (R^2 - Y)\,\mathrm{d}T + \eta\,\mathrm{d}B$.

Proof. $\mathrm{d}r = \varepsilon_2^{1/3}\,\mathrm{d}R$; the drift $(br^2 - ay)\,\mathrm{d}t = \varepsilon_2^{2/3}(bR^2 - aY)\cdot\varepsilon_2^{-1/3}\,\mathrm{d}T = \varepsilon_2^{1/3}(bR^2 - aY)\,\mathrm{d}T$; and the noise $\sigma\,\mathrm{d}W_t = \sigma(\mathrm{d}t)^{1/2}\hat Z = \sigma(\varepsilon_2^{-1/3}\,\mathrm{d}T)^{1/2}\hat Z = \sigma\,\varepsilon_2^{-1/6}\,\mathrm{d}B_T$. Dividing $\mathrm{d}r = \varepsilon_2^{1/3}\,\mathrm{d}R$ through by $\varepsilon_2^{1/3}$: $\mathrm{d}R = (bR^2 - aY)\,\mathrm{d}T + \sigma\,\varepsilon_2^{-1/2}\,\mathrm{d}B_T$. Similarly $\varepsilon_2^{2/3}\,\mathrm{d}Y = -\varepsilon_2 c\,\varepsilon_2^{-1/3}\,\mathrm{d}T$ gives $\mathrm{d}Y = -c\,\mathrm{d}T$. ∎

Remark (scales). The chart inner scales $R \sim \eta^{2/3}$, $Y \sim \eta^{4/3}$ blow down, via (2) and $\eta = \sigma/\sqrt{\varepsilon_2}$, to the *physical* noise-induced peel-off scales

$$r_{\text{peel}} \sim \varepsilon_2^{1/3}\eta^{2/3} = \sigma^{2/3}, \qquad y_{\text{peel}} \sim \varepsilon_2^{2/3}\eta^{4/3} = \sigma^{4/3},$$

independent of ε_2 — the noise-induced jump scales — while the typical (canard) escape sits at $y = \varepsilon_2^{2/3} a_1$.

2 The blow-down theorem

Blowing down the chart conclusions (the reduction, the shooting characterisation, the instanton rate, and the composition propagation, all θ-uniform) through (2) and Proposition 1 gives the physical statement.

Theorem 1 (Path A, physical variables)**.** *For (1), modulo the per-chart Berglund–Gentz tubes and the JKK deterministic backbone:*

(i) (critical noise) *escape across the fold becomes likely at*

$$\boxed{\sigma_*^A = 2\sqrt{\pi}\,\sqrt{\varepsilon_2}\,G}, \qquad G = \begin{cases} \sqrt{a(\theta_*)c(\theta_*)/b(\theta_*)}, & \alpha = 2 \ (\textit{frozen } \theta), \\ \sqrt{\langle a\rangle\langle c\rangle/\langle b\rangle}, & \alpha = 1 \ (\textit{averaged}), \end{cases}$$

uniformly in θ;

(ii) (inner exit measure) *the peel-off level is $y_{\text{peel}} = \varepsilon_2^{2/3} Y_{\text{node}}$ with $Y_{\text{node}} \overset{d}{=} \mathrm{TW}_\beta$, $\beta = 4\varepsilon_2/\sigma^2$; equivalently the early-escape probability is $\mathbb{P}_{\text{early}} = 1 - F_\beta(0)$;*

(iii) (early-escape rate) *as $\sigma/\sqrt{\varepsilon_2} \to 0$, $\mathbb{P}_{\text{early}} = \exp\big(-c\,\varepsilon_2/\sigma^2\,(1+o(1))\big)$, $c = 5.4439\ldots$*

All three depend on (σ, ε_2) only through $\eta = \sigma/\sqrt{\varepsilon_2}$ at leading order.

Proof. By Proposition 1 the K_2 dynamics is the canonical inner Riccati with $\eta = \sigma/\sqrt{\varepsilon_2}$. (ii) is the reduction theorem ($Y_{\text{node}} \overset{d}{=} \mathrm{TW}_\beta$, $\beta = 4/\eta^2 = 4\varepsilon_2/\sigma^2$) blown down by $y = \varepsilon_2^{2/3} Y$; (iii) is the instanton rate $e^{-c/\eta^2} = e^{-c\varepsilon_2/\sigma^2}$. For (i): escape is order-one when the integrated inner hazard $H = \eta^2/4\pi$ is order one, i.e. $\eta_* = 2\sqrt{\pi}$, so $\sigma_* = 2\sqrt{\pi}\sqrt{\varepsilon_2}$ in the canonical chart; restoring a, b, c (the linear rescaling that canonicalises $bR^2 - aY$ with sweep rate c) multiplies by the geometry factor G. The composition-propagation theorem carries these inner laws through the entry (K_1) and exit (K_3) charts with θ-uniform, sub-dominant Gaussian corrections (the fold renewal annihilating the entry deviation); the deterministic geometry is JKK Theorem 3.2, whose α-dichotomy (frozen vs. averaged θ) gives the two cases of G. Every estimate is θ-uniform by the θ-uniformity lemma. $\square$

Verification of the collapse. Simulating the *physical* system (1) ($a = b = c = 1$) through the fold and rescaling $Y = y_{\text{peel}}/\varepsilon_2^{2/3}$, the early-escape probability and the peel-off law depend only on $\eta = \sigma/\sqrt{\varepsilon_2}$:

$$\begin{array}{lll} \eta = 1.5: & (\varepsilon_2, \sigma) = (0.05, 0.335): \ \mathbb{P}_{\text{early}} = 0.117,\ \langle Y\rangle = -1.20 & (0.20, 0.671): \ 0.113,\ -1.21 \\ \eta = 1.0: & (\varepsilon_2, \sigma) = (0.05, 0.224): \ \mathbb{P}_{\text{early}} = 0.011,\ \langle Y\rangle = -1.60 & (0.20, 0.447): \ 0.010,\ -1.60 \end{array}$$

a clean collapse across a factor 4 in ε_2 (`blowdown.py`).

3 Uniformity and the assembled statement

Uniformity in $(\varepsilon_1, \varepsilon_2)$. The ε_2-dependence enters only through the blow-up radius $\rho = \varepsilon_2^{1/3}$ and the chart noise $\eta = \sigma/\sqrt{\varepsilon_2}$ (Proposition 1), so the chart estimates—θ-uniform and ε_2-uniform once written in η—transfer verbatim. The $\varepsilon_1 = \varepsilon^\alpha$ dependence enters only through the angular drift $\dot\theta = \varepsilon_1$: for $\alpha = 2$ the angle is frozen on the passage timescale and G is local; for $\alpha = 1$ the angle is fast and Khasminskii averaging replaces a, b, c by $\langle a\rangle, \langle b\rangle, \langle c\rangle$ with $O(1/\varepsilon_1)$ corrections. This is exactly JKK Theorem 3.2's deterministic α-dichotomy, now carrying the noise.

The assembled Path A. Theorem 1 is the capstone of the chain: *Cole–Hopf* (exact Itô) $\Rightarrow$ *reduction* to the stochastic Airy operator $\Rightarrow$ *shooting characterisation* ($Y_{\text{node}} \overset{d}{=} -\Lambda_0 \overset{d}{=} \mathrm{TW}_\beta$, proved) $\Rightarrow$ *instanton* rate $c = 5.4439$ (proved) $\Rightarrow$ *composition propagation* (θ-uniform, fold renewal, proved) $\Rightarrow$ *blow-down* (this note) to the physical critical-noise, exit-measure and rate laws. The folded-limit-cycle canard escape is thereby placed in the Tracy–Widom / KPZ edge universality class, in the original variables.

4 Honest scope

The blow-down is rigorous *given* the chart results: Proposition 1 is an exact change of variables, the scale blow-downs are algebra, and the assembly is the composition theorem plus JKK's deterministic backbone. The remaining softness is inherited, not introduced: the per-chart K_1, K_3 tubes are cited Berglund–Gentz (standard for hyperbolic charts), and Channel B is scaling-level. What this note adds is the final desingularisation—turning the $O(1)$ chart statements into the physical-(σ, ε_2) theorem—which was the last named structural item of Path A.

References

[1] M. Krupa, P. Szmolyan, *Extending GSPT to nonhyperbolic points*, SIAM J. Math. Anal. **33** (2001) 286–314 (the fold blow-up).

[2] S. Jelbart, C. Kuehn, N. Kuntz, arXiv:2208.01361 (2024), Theorem 3.2.

[3] N. Berglund, B. Gentz, *Noise-Induced Phenomena in Slow–Fast Dynamical Systems*, Springer (2006).

[4] Companion notes: `ShootingCharacterisation_proof`, `InstantonPrinciple_proof`, `CompositionPropagation_proof`, `ChannelB_SNIC_proof`, `PathA_endgame_writeup`.